

Is newer always better?

Transformers against CNNs in multi-class one dimensional signal classification

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Abstract

Automated classification of electrocardiogram (ECG) heartbeats is essential for the early diagnosis and management of a multitude of heart conditions, such as but not limited to cardiac arrhythmias. However, severe class imbalance common in medical datasets and subtle morphological distinctions across arrhythmia types often hinder predictive performance in minority classes. In this work, an end-to-end Transformer-based classification framework was developed and evaluated on the benchmark MIT-BIH Arrhythmia Dataset, set up in comparison as a newer approach to existing benchmarks that are using older state-of-the-art architectures, such as Convolutional Neural Networks. The architecture features a 4-layer Transformer Encoder employing multi-head self-attention ($n_{head}=4$, $d_{model}=64$), sinusoidal positional encodings, and a learnable classification token to model temporal heartbeat dynamics. To mitigate extreme label disproportion, training incorporated class-weighted cross-entropy loss alongside AdamW optimization, cosine annealing learning rate schedules, gradient clipping, and mixed-precision computation. The model converged via early stopping at epoch 94, yielding an overall test accuracy of 99% and a test macro F1-score of 0.9313. Notably, the balanced loss scheme yielded high sensitivity across minority pathologies - demonstrating the efficacy of self-attention mechanisms for reliable arrhythmia detection in clinically imbalanced settings.

Introduction

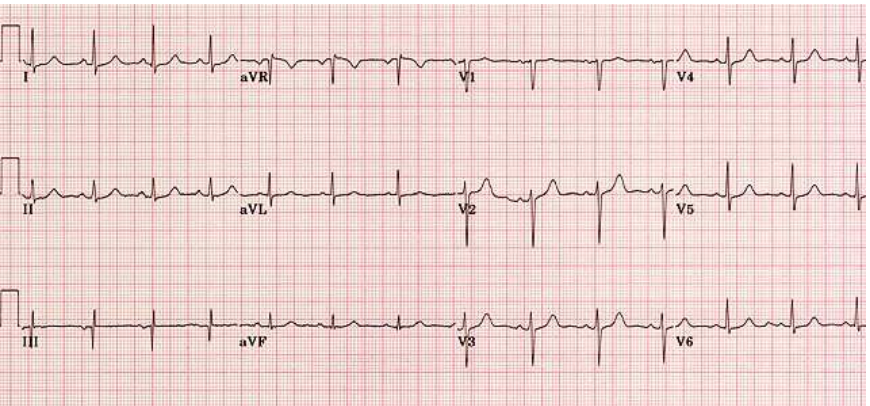
Cardiovascular diseases remain the leading cause of global mortality, necessitating timely and precise identification of abnormal cardiac rhythms. The electrocardiogram (ECG) is the foundational non-invasive clinical diagnostic tool for monitoring electrophysiological activity, capturing subtle morphology shifts that signal underlying heart conditions. To alleviate the burden and subjectivity of manual Holter reading, automated beat-level classification has long relied on established 1D signal-processing baselines, predominantly one-dimensional

Convolutional Neural Networks and recurrent architectures. While these classic state-of-the-art models excel at extracting localized waveform features, their bounded receptive fields often restrict their ability to model long-range temporal dependencies across the full cardiac cycle. Moreover, medical datasets inherently suffer from severe class imbalance, where conventional CNN-based pipelines tend to optimize for overall accuracy at the expense of rare, life-threatening arrhythmia classes.

In response to these limitations, recent advancements in deep learning have shifted toward exploring newer state-of-the-art architectures, specifically self-attention mechanisms; to test whether global sequence modeling can marginally improve upon established 1D-CNN baselines. This work investigates an end-to-end 1D Transformer Encoder designed to process raw ECG heartbeats without inductive convolutional biases, utilizing sinusoidal positional encodings and a dedicated classification token to capture global morphology and cross-temporal interactions. Evaluated on the benchmark MIT-BIH Arrhythmia Dataset alongside class-weighted optimization, this study examines whether modern Transformer designs can push performance boundaries by achieving high recall across scarce, clinically critical categories while retaining competitive global classification accuracy.

ECG Dataset

The evaluation is conducted on the benchmark Physionet MIT-BIH Arrhythmia Dataset[1] preprocessed in accordance with the Association for the Advancement of Medical Instrumentation (AAMI EC57) standard. In this formulation, continuous two-lead electrocardiogram recordings



are segmented into individual, *Illustration 1: An example of an ECG signal. Source:*

isolated heartbeat cycles, <https://oxfordmedicaleducation.com/ecgs/ecg-examples/>

centered around their detected R-peaks, normalized, and uniformly resampled to a frequency of 125 Hz. Each sample forms a fixed-length 1D sequence of 186 time steps paired with a ground-truth categorical label. The benchmark groups all annotated heartbeat morphologies into five mutually exclusive clinical categories: Non-Ectopic beats (Class 0), which encompass normal sinus beats and bundle branch blocks; Supraventricular ectopic beats (Class 1), including atrial and nodal premature contractions; Ventricular ectopic beats (Class 2), containing premature ventricular contractions and ventricular escape beats; Fusion beats (Class 3), representing hybrid activation waveforms between ventricular and normal conduction; and Unknown beats (Class 4), covering paced beats and unclassifiable artifacts. Across the dataset, there is acute class imbalance: majority Non-Ectopic beats constitute over 82% of the total samples, while critical minority classes such as Supraventricular and Fusion beats represent approximately 2.5% and less than 1% of the distribution, respectively. This

pronounced disproportion, illustrated in the accompanying class distribution plot, highlights the necessity for evaluation strategies that prevent classification models from overlooking rare cardiac events. This dataset is being accessed from an [existing Kaggle repository](#) in which the waveform data of the ECGs has already been processed and normalised.

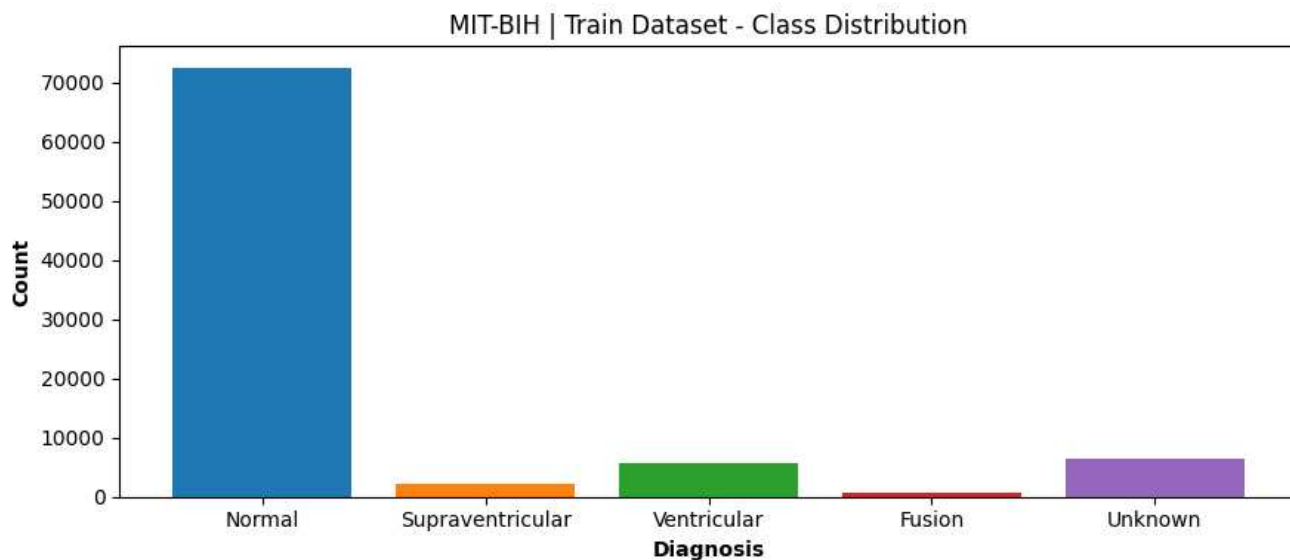


Figure 1: Class distribution of MIT-BIH training data.

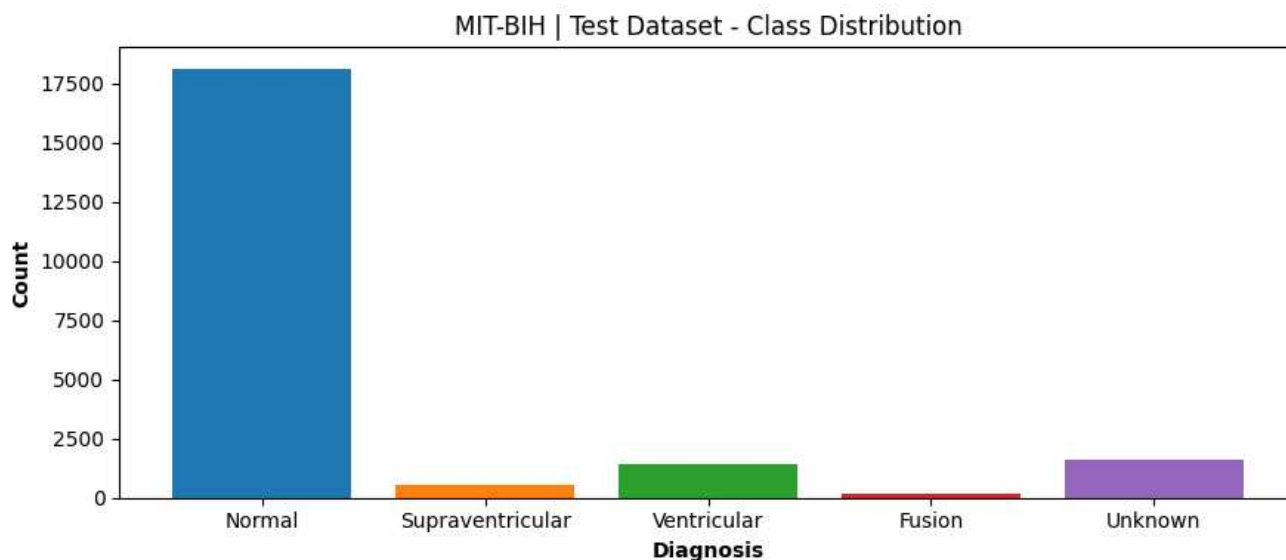


Figure 2: Class distribution of MIT-BIH test data.

1D Transformers

Originally introduced by Vaswani et al.[2] for sequence-to-sequence translation, the Transformer architecture has expanded far beyond natural language processing into computer vision (Dosovitskiy et al.) [3] and one-dimensional temporal signal analysis (Zerveas et al.) [4]. Unlike traditional Convolutional Neural Networks that rely on localized kernel operations, or Recurrent Neural Networks that process steps sequentially and struggle

with vanishing gradients, a 1D Transformer models an entire time series simultaneously via self-attention. When applied to raw 1D sequential data, continuous scalar measurements across L time steps are mapped into an embedding space of dimension d_{model} , either through a point-wise linear projection or a 1D patch embedding layer-producing an input matrix $X \in \mathbb{R}^{L \times d}$. Because the standard self-attention operation is permutation-invariant, it contains no inherent awareness of temporal ordering. To preserve sequential structure, deterministic sinusoidal functions or learned vector embeddings are added directly to the projected tokens:

$$PE_{(pos, 2i)} = \sin\left(\frac{pos}{10000^{2i/d_{model}}}\right), \quad PE_{(pos, 2i+1)} = \cos\left(\frac{pos}{10000^{2i/d_{model}}}\right)$$

Additionally, following classification paradigms established by BERT (Devlin et al.) [5] and Vision Transformers, a learnable classification token ([CLS]) is prepended to the embedded sequence. As the sequence passes through successive layers, this token interacts globally with every time step, serving as an aggregate summary vector of the entire signal for downstream prediction tasks. The core computational mechanism of the 1D Transformer Encoder is the Scaled Dot-Product Multi-Head Self-Attention module. The token representations are projected into Query Q , Key K , and Value V matrices via learned linear weights. The attention weight between any two temporal steps indicates their mutual relevance across the sequence, computed as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

By running multiple attention heads in parallel, the network captures complementary temporal patterns simultaneously across varying time scales. Each encoder block couples MHSA with a position-wise feed-forward network, residual connections, and layer normalization to maintain stable gradient flow throughout training. For classification objectives, the processed representation corresponding to the [CLS] token is extracted from the final encoder layer and fed into a multi-layer perceptron (MLP) head to generate output logits.

Experimental Setup

The dataset, its preprocessing and train/test split are all iterated upon as seen on the Kaggle dataset, with the ECG signals already normalised and collated to an equal length.

The baseline CNN [6] architecture is designed for beat-level classification by taking the preprocessed individual heartbeats as direct sequential inputs. It employs one-dimensional temporal convolutions across all feature-extraction stages, with each convolutional layer utilizing 32 filters of kernel size 5. Downsampling

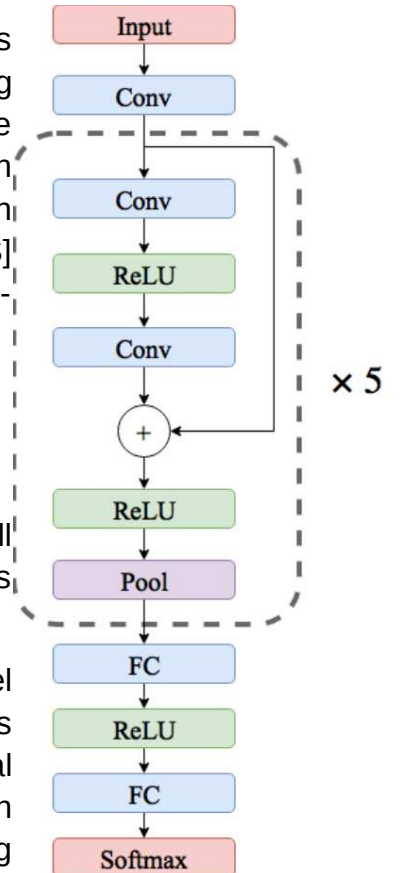


Figure 3: CNN architecture of the baseline study [6].

is carried out uniformly using max pooling with a pool size of 5 and a stride of 2. The core backbone is composed of five residual blocks, wherein each block integrates two convolutional layers, two ReLU activation functions, a residual skip connection, and a pooling layer. Following the residual feature extraction stages, the network terminates in two fully connected layers of 32 units each, culminating in a softmax output layer that yields the predicted class probabilities across a total depth of 13 parameter-bearing weight layers.

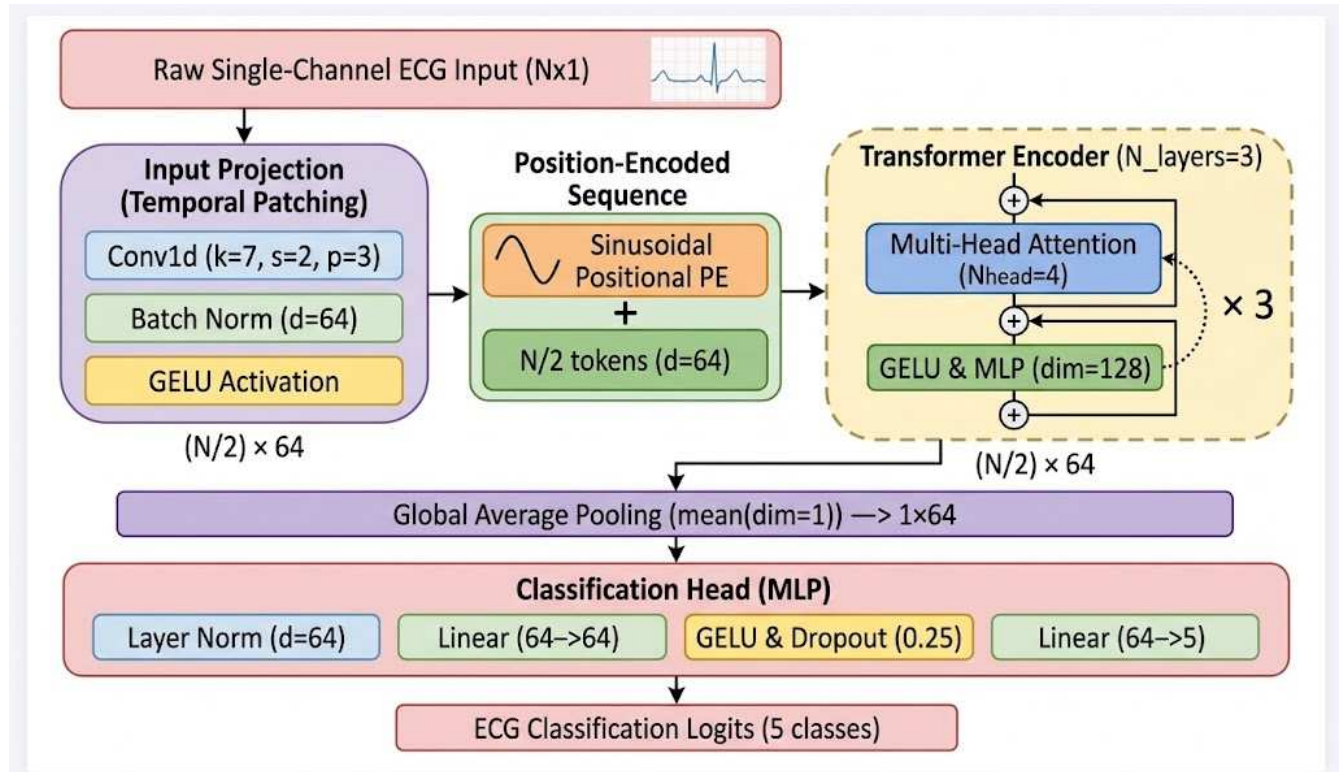


Figure 4: Proposed Transformer network.

The revised model architecture transitions from a purely linear token projection to a hybrid convolutional-Transformer design tailored for 1D ECG signals. Rather than embedding each scalar time step individually, the raw single-channel heartbeat sequence is initially processed through a 1D convolutional stem featuring a kernel size of 7, a stride of 2, and a padding of 3, projecting the input directly into a 64-dimensional latent space while halving the sequence length. This layer is followed by 1D batch normalization and a GELU non-linearity to stabilize activations and introduce localized morphological priors prior to self-attention modeling. Deterministic sinusoidal positional encodings are then added to the downsampled sequence to preserve the temporal ordering of cardiac phases across the temporal dimension. Sequence encoding is carried out by a streamlined, 3-layer Transformer Encoder using 4 attention heads, a model dimension of 64, an internal feed-forward expansion of 128, and a dropout rate of 0.25. In contrast to standard classification-token paradigms, temporal aggregation is performed via global average pooling across all encoded token states. This pooled vector passes into a multi-layer perceptron head consisting of layer normalization, a

64-unit hidden linear layer with GELU activation and dropout, and a final linear layer outputting logits over the five target classes. Training is configured under full seed reproducibility and optimized with an inverse-frequency class-weighted cross-entropy loss to counteract dataset imbalance. The network is trained using AdamW and a cosine annealing learning rate schedule across 100 epochs, utilizing automatic mixed precision and gradient norm clipping. Augmentation was only applied to the minority classes to reduce overfitting to the majority class, methods including Gaussian noise injection ($\mu = 0.0$, $\sigma = 0.015$) to emulate baseline sensor interference, random amplitude scaling by a factor between 0.85 and 1.15 to simulate inter-patient voltage variations, temporal shifting within ± 10 samples with zero-padding to account for misalignments in R-peak centering, and random cutout masking of continuous segments spanning 5 to 15 time steps to encourage robustness against localized signal artifacts and lead dropouts. Each perturbation was applied stochastically during training with independent activation probabilities, leaving normal heartbeat waveforms unaltered to prevent distorting baseline morphology.

Results

The model achieves optimal performance, reaching an overall test classification accuracy of 99% and a macro-averaged F1 of 0.9313 across all 21,892 test samples. In the following figures containing confusion matrices comparing both studies, it is evident that the transformer architecture edges over the traditional CNN in most cases. A noteworthy disclaimer to be added here is the distribution of the data in which these 2 models are being tested with. The CNN paper simply states that it created the test set from 819 samples from each class that weren't to be used in the training data. The test set as set-up in the Kaggle repository doesn't match this specific setup, though when trying to run experiments mirroring this training setup for the transformer the results were unsatisfactory. The full training set was used for training the transformer. This does reflect a difference in class quantities in the test data, but it does provide a fuller context for transformer's performance. A balanced subset was created by undersampling the other classes to the minimum of the class with the lowest count of samples.

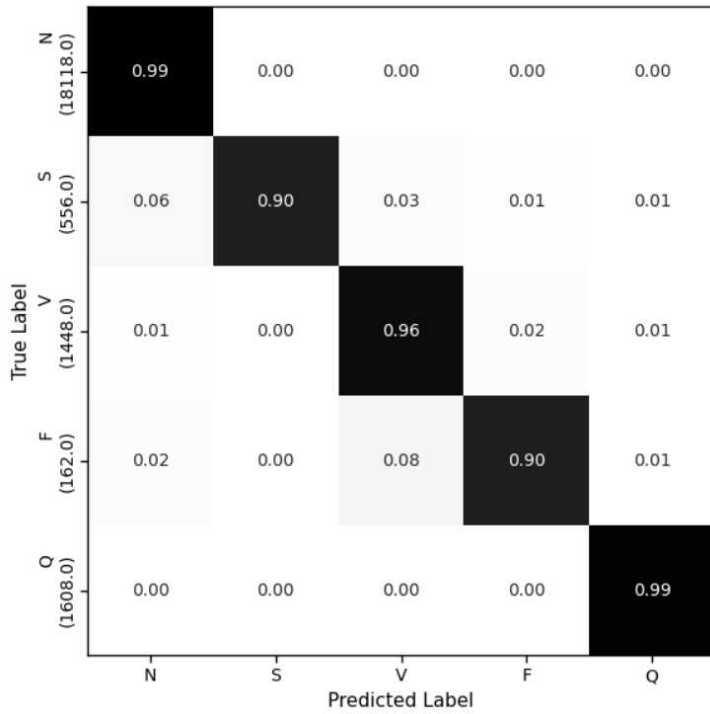


Table 1: Confusion Matrix - Transformer, full test set.

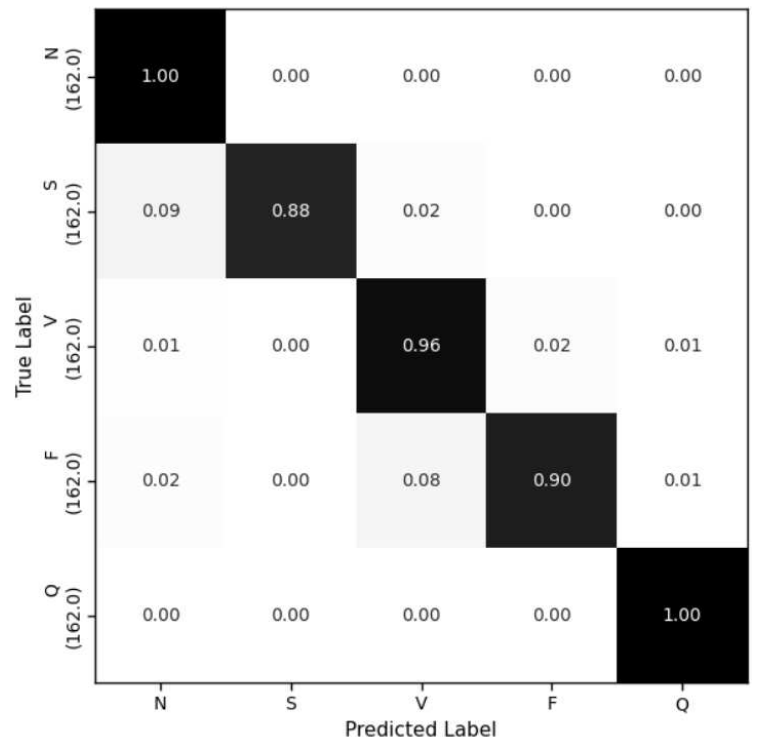


Table 2: Confusion Matrix - Transformer, balanced test set.

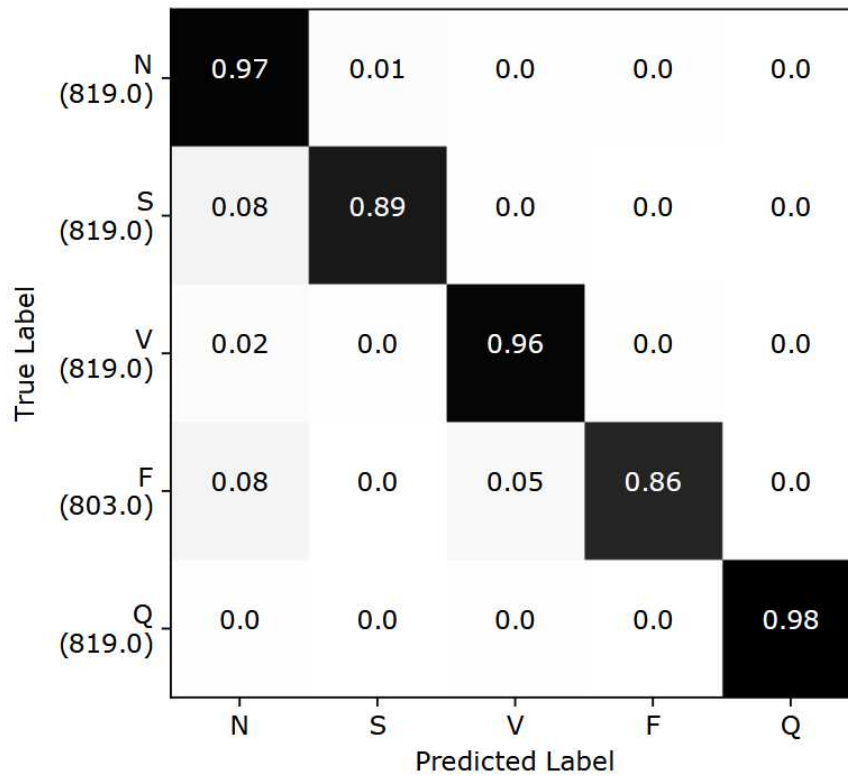


Table 3: Confusion Matrix - CNN, balanced test set. Note the quantity of instances in each class compared to Table 2. [4]

Conclusion

We can safely conclude that, given sufficient data scale alongside tailored tuning and domain-specific augmentations, transformer architectures can effectively leverage the extensive representation spaces found in medical signal datasets to outperform traditional convolutional neural networks. While extreme class imbalance has historically posed severe convergence and generalization challenges in biomedical time-series analysis, pairing targeted data augmentation for underrepresented classes with frequency-weighted optimization demonstrates that self-attention mechanisms capture subtle, long-range morphological dependencies that standard local kernels miss. Furthermore, with additional hyperparameter refinement and advanced regularizations, these models possess clear headroom for even higher discriminative fidelity. It is therefore worthwhile to revisit previous benchmarks and experimental setups where CNNs were established as the definitive standard, evaluating whether modern, well-regularized transformer frameworks can yield systematic, incremental gains across diverse clinical diagnostics.

Sources

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